

Smart Meter Fraud Detection & Anomaly Analysis Report

1. Abstract

Smart meter fraud and electricity theft pose significant economic losses for utility providers. This study leverages machine learning algorithms to detect fraudulent activities in energy consumption patterns. Using Isolation Forest, Random Forest, and XGBoost, we identify high-risk customers, analyze temporal trends, and map fraud-prone locations. The findings suggest that XGBoost outperforms other models in accuracy and recall. We provide actionable insights to utility companies to mitigate fraud and improve revenue recovery.

2. Introduction

Electricity theft via smart meter tampering is a growing problem in South Africa. Fraudulent users manipulate meters to reduce bill amounts, leading to significant losses. This study aims to detect anomalies and fraudulent activities using machine learning models.

3. Problem Statement

- Economic loss due to fraud (estimated billions annually in SA)
 - Difficulties in manual fraud detection
 - Need for automated, AI-driven fraud detection systems
-

4. Data Collection & Preprocessing

Dataset Columns:

- Meter ID, Timestamp, Province, City, Latitude, Longitude
- Energy Consumption (kWh), Solar Generation (kWh), Voltage, Frequency, Power Factor, Fraud Label
- Load Shedding Impact Considered

Preprocessing Steps:

- Timestamp conversion → Extract hour, day, month, weekday
 - Feature Engineering → Calculate Power Efficiency, Consumption per Voltage
 - Scaling & Normalization → Standardized for ML models
-

5. Machine Learning Models & Evaluation

We implemented multiple models:

- Isolation Forest → Unsupervised anomaly detection
- Random Forest → Supervised classification
- XGBoost → Advanced tree-based model for fraud classification
- Logistic Regression → Baseline model

Performance Metrics:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
XGBoost	98%	93%	88%	90%	97%
Random Forest	96%	89%	82%	85%	94%
Isolation Forest	88%	70%	91%	79%	90%
Logistic Regression	92%	85%	79%	82%	92%

Best Model: XGBoost due to high accuracy and fraud detection capability.

6. Results & Insights

Trend Analysis

- Fraudulent activities peak between 12 AM - 5 AM
- Fraudsters exhibit sudden drops in consumption, indicating meter tampering
- Johannesburg, Cape Town, Durban → Highest fraud risk cities

Fraud Risk Map (Geospatial Analysis)

- Used GeoPandas & Shapefiles for fraud mapping
 - High-risk areas were visualized on a South African map
 - Targeted inspections can be prioritized in red-zone areas
-

7. Actionable Recommendations

For Utility Companies

- Deploy XGBoost-based fraud detection system in real-time
- Increase meter inspections in high-risk areas
- Implement automated alerts for anomaly detection

For Policy Makers

- Strengthen laws against electricity theft
- Introduce fines and penalties for fraudsters

For Consumers

- Awareness campaigns on the risks of illegal connections
 - Incentivize solar adoption for honest users
-

8. Conclusion & Future Work

Conclusion:

This study demonstrated that machine learning can effectively detect fraud in smart meters. XGBoost outperforms other models and can be deployed for real-time monitoring.

Future Work:

- Real-time fraud detection API (FastAPI) for fraud alerts
- Incorporate deep learning models (LSTMs, CNNs) for pattern recognition
- Integration with IoT-based smart meters for real-time prevention